MATH 130 · UNIT 3C

Right Triangle Applications

Concepts, worked examples, student practice, and synthesis

55 minutes · Offline capable

Today’s Learning Roadmap

1. model one-triangle applications with SOH-CAH-TOA
2. apply elevation, depression, and motion components
3. solve complete and two-triangle application problems

Predict: model one-triangle applications with SOH-CAH-TOA

PREDICT FIRST

What must be labeled before selecting a trig ratio?

Commit to a response before discussion.

One right triangle: read, draw, label, solve, check

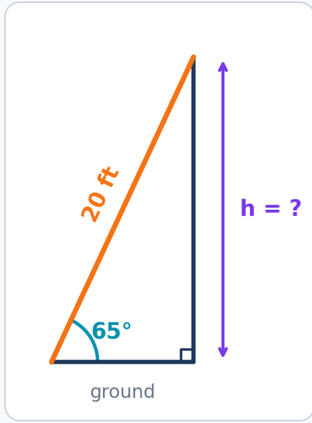
CONCEPT

- Read, draw, label, choose a ratio, solve, and check.
- Inverse trig finds an angle from a ratio.

Model the Skill: model one-triangle applications with SOH-CAH-TOA

ANNOTATED DIAGRAM

- A 20 ft ladder makes a 65° angle with the ground. Find the height.
- $\sin 65^\circ = h/20$.
- $h = 20 \sin 65^\circ \approx 18.13$ ft.
- The height is less than the 20 ft ladder, as it must be.



Right triangle ladder diagram with a 20 foot ladder and 65 degree ground angle

You Try: model one-triangle applications with SOH-CAH-TOA

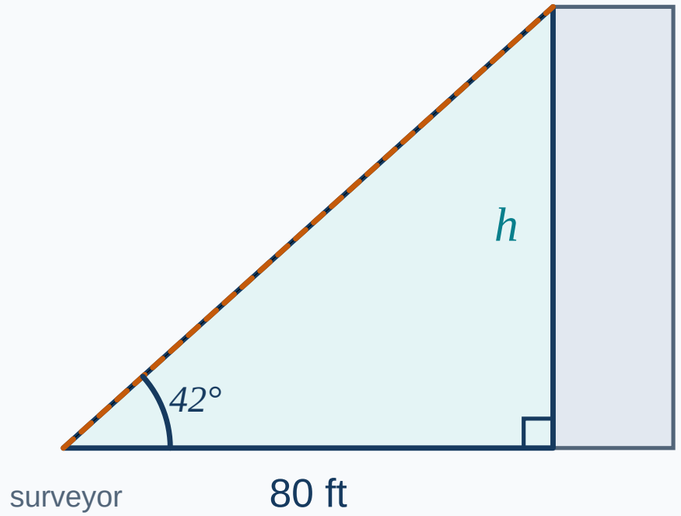
STUDENT PRACTICE

A surveyor stands 80 ft from a building at 42° elevation. Find the height.

Reveal solution

FEEDBACK AND SOLUTION

$h = 80 \tan 42^\circ \approx 72.0$ ft.



The building height h is opposite the 42° angle.

Check and Diagnose: model one-triangle applications with SOH-CAH-TOA

ERROR ANALYSIS

- Calculator mode must be degrees.
- A leg must be shorter than the hypotenuse.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Make It Stick: model one-triangle applications with SOH-CAH-TOA

PACING CHECKPOINT

State the six-step word-problem strategy.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Predict: apply elevation, depression, and motion components

PREDICT FIRST

Why are an angle of depression and its matching angle of elevation equal?

Commit to a response before discussion.

Elevation, depression and components

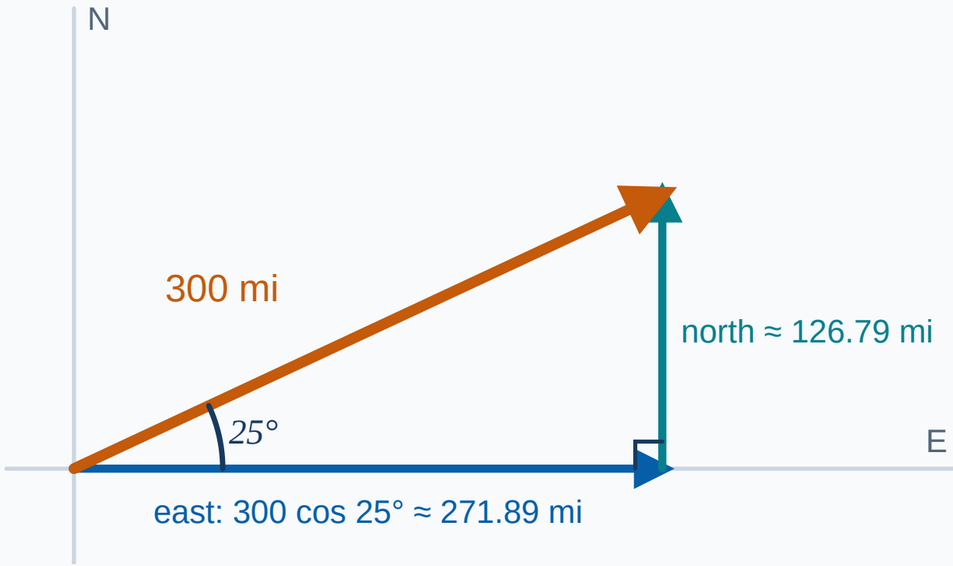
FORMULA FOCUS

- Both are measured from parallel horizontal lines.
- Resolve motion with $d_x = d\cos\theta$ and $d_y = d\sin\theta$.

Model the Skill: apply elevation, depression, and motion components

A vehicle travels 300 miles at 25° north of east. Resolve its displacement.

1. East component: $300 \cos 25^\circ \approx 271.89$ mi.
2. North component: $300 \sin 25^\circ \approx 126.79$ mi.
3. $\sqrt{271.89^2 + 126.79^2} \approx 300$ verifies the components.



The two components form the legs of a right triangle.

You Try: apply elevation, depression, and motion components

STUDENT PRACTICE

A plane flies 450 miles at 35° north of east. Find its east and north components.

Reveal solution

FEEDBACK AND SOLUTION

East: $450 \cos 35^\circ \approx 368.62$ mi; north: $450 \sin 35^\circ \approx 258.11$ mi.

Check and Diagnose: apply elevation, depression, and motion components

ERROR ANALYSIS

- Angles are measured from horizontal unless stated otherwise.
- Use total distance before resolving components.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Make It Stick: apply elevation, depression, and motion components

PACING CHECKPOINT

Explain how a horizontal reference line organizes both elevation and component problems.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Predict: solve complete and two-triangle application problems

PREDICT FIRST

What information is shared by two right triangles aimed at the same height?

Commit to a response before discussion.

Solving complete and two-triangle problems

CONCEPT

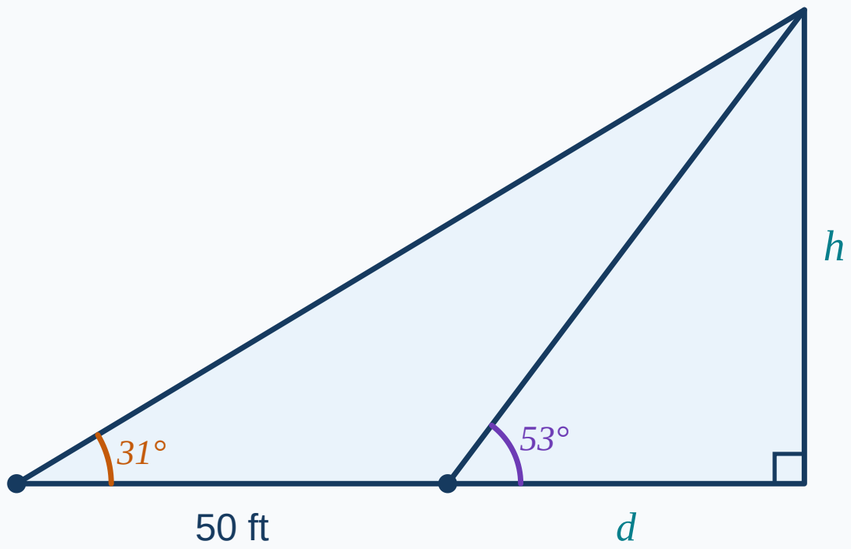
- To solve a right triangle, find all missing sides and angles.
- Two-triangle problems share a height and use related horizontal distances.

Source: 12, 13

Model the Skill: solve complete and two-triangle application problems

Two tower observations are 50 ft apart with elevation angles 53° and 31°. Find the closer distance d and height h.

1. $h = d\tan 53^\circ = (d + 50)\tan 31^\circ$.
2. $d \approx 41.37$ ft and $h \approx 54.90$ ft.
3. Using the farther distance 91.37 ft reproduces the same height.



Drawn to scale. The near point is d feet from the base.

You Try: solve complete and two-triangle application problems

STUDENT PRACTICE

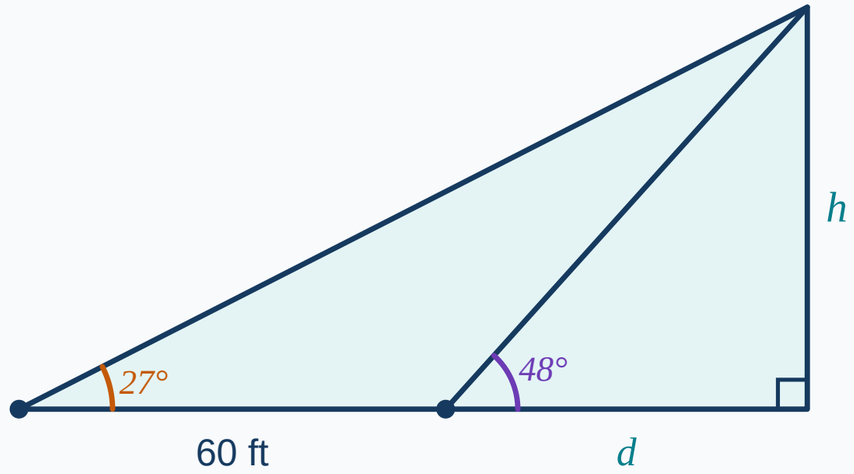
Two points are 60 ft apart. From the far point the angle of elevation to the top of a building is 27°; from the near point it is 48°. Set up both equations, then find d and h.

Reveal solution

FEEDBACK AND SOLUTION

$\tan 48^\circ = h/d$ and $\tan 27^\circ = h/(d+60)$, so

$$d = \frac{60\tan 27^\circ}{\tan 48^\circ - \tan 27^\circ} \approx 50.86 \text{ ft and } h \approx 56.49 \text{ ft.}$$



Givens only: find d and h.

Check and Diagnose: solve complete and two-triangle application problems

ERROR ANALYSIS

- Keep d and $d + 50$ attached to the correct observation points.
- Show the algebraic distribution step before dividing.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Make It Stick: solve complete and two-triangle application problems

PACING CHECKPOINT

Write the shared-height equation that connects the two triangles.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Session Summary and Exit Ticket

1. State the most important idea from today without looking at your notes.
2. Complete one representative setup and identify the step most likely to cause an error.
3. Write one question that should be answered before the next assessment.

Exit response

Answer in complete mathematical sentences and identify one remaining question.

Reveal review guidance

FEEDBACK AND SOLUTION

Use the objective roadmap to identify any skill that still needs deliberate practice.